
System Requirements & Hazards Analysis

Robotic Intervention for Pollution and Plastic Litter Extraction (RIPPLE)

MECHTRON 4TB6: Mechatronics Capstone Design Process

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Group 14

November 18th, 2025

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Purpose of Document

The purpose of this document is to evaluate the hazards that can arise from a remote-controlled surface-water trash collecting system in an attempt to mitigate the chances of system failure or other unintended negative consequences. The document will further highlight a list of resulting safety requirements from this analysis. In addition to that, the document will define the high-level system requirements that will be used to guide the design process and for end product verification purposes.

Scope of Project

The system is intended to be used by a trained adult familiar with the controls in a pond or other small body of water that has accumulated trash on its surface in calm, non-precipitating weather. The project is intended to collect the floating trash as directed by its operator.

System Definition

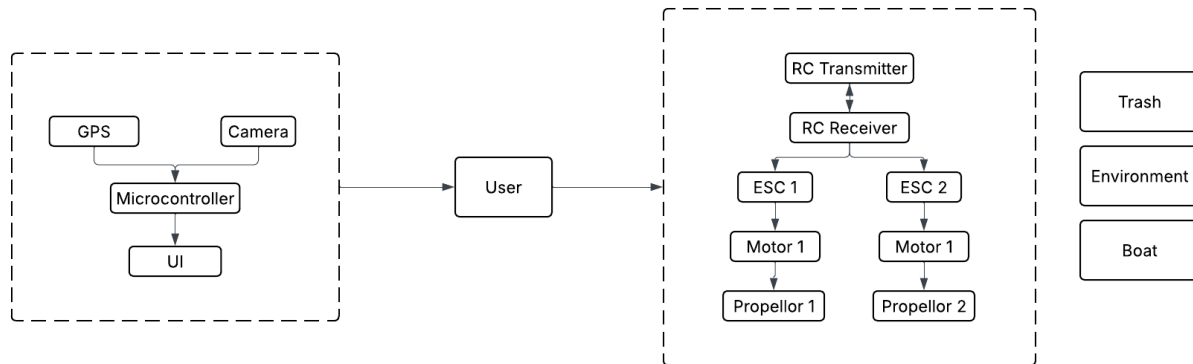
Definitions

Table 1: Definitions

Word/Acronym	Definition
Boat Hull	Outer body of the TCM (not including motors and microcontroller)
ESC	Electronic Speed Controller
HID	Hazard Identifier Number
RC	Remote/Radio Controller
RID	Requirement Identifier Number
SSR	Safety & Security Requirement
System	Overall system including: TCM, RC and UI
TCM	Trash Collection Mechanism. This includes the “hull” of the boat, the net used to contain the collected trash and all associated electronic components on the boat.
Trash	Floating trash on the surface of the pond that is within the size constraints of the TCM.
UI	User Interface
UM	User’s Manual
RPM	Revolutions Per Minute

System Diagram Showing Boundaries

Figure 1: System Diagram Showing Boundaries



“The System” as described in this document includes two major sub-systems that both interact with the user. Sub-system one includes the GPS and Camera modules controlled by a microcontroller whose main functions are to provide feedback of the system’s state to the user via a UI. The second sub-system’s primary goal is to transform user inputs into desired movement, achieved through an RC connected to the boat’s propellers. Other important elements of the project that fall outside the controllable boundaries of the system include the trash the project wishes to collect, the system’s environment, and the Boat Hull.

Monitored & Controlled Variables

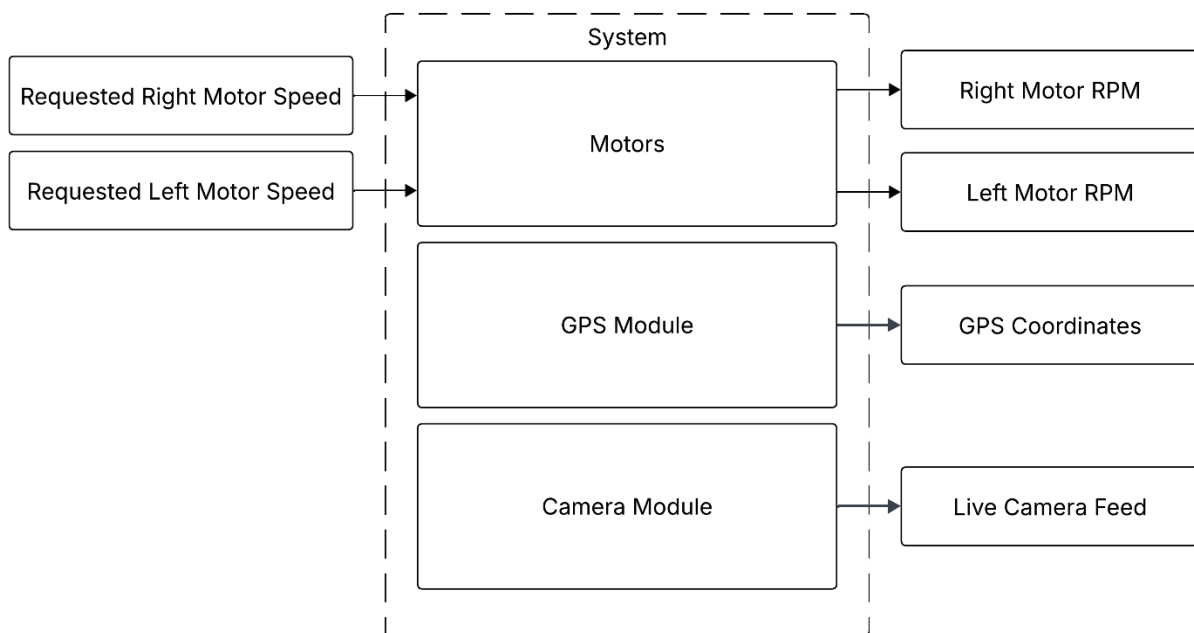
Table 2: Monitored Variables

Monitored Variables (System Inputs)	Description
Requested Right Motor Speed	This is the right motor speed that the user will request by moving the joystick on the RC.
Requested Left Motor Speed	This is the left motor speed that the user will request by moving the joystick on the RC.

Table 3: Controlled Variables

Controlled Variables (System Outputs)	Description
Right Motor RPM	This is the actual right motor speed that will turn the right propeller.
Left Motor RPM	This is the actual left motor speed that will turn the left propeller.
GPS Coordinates	These are the x and y coordinates of the system, that will be updated in real-time and be displayed on the UI.
Live Camera Feed	This is the live camera feed that will be updated in real-time and be displayed on the UI.

Figure 2: Context Diagram



System Requirements & Hazards Analysis

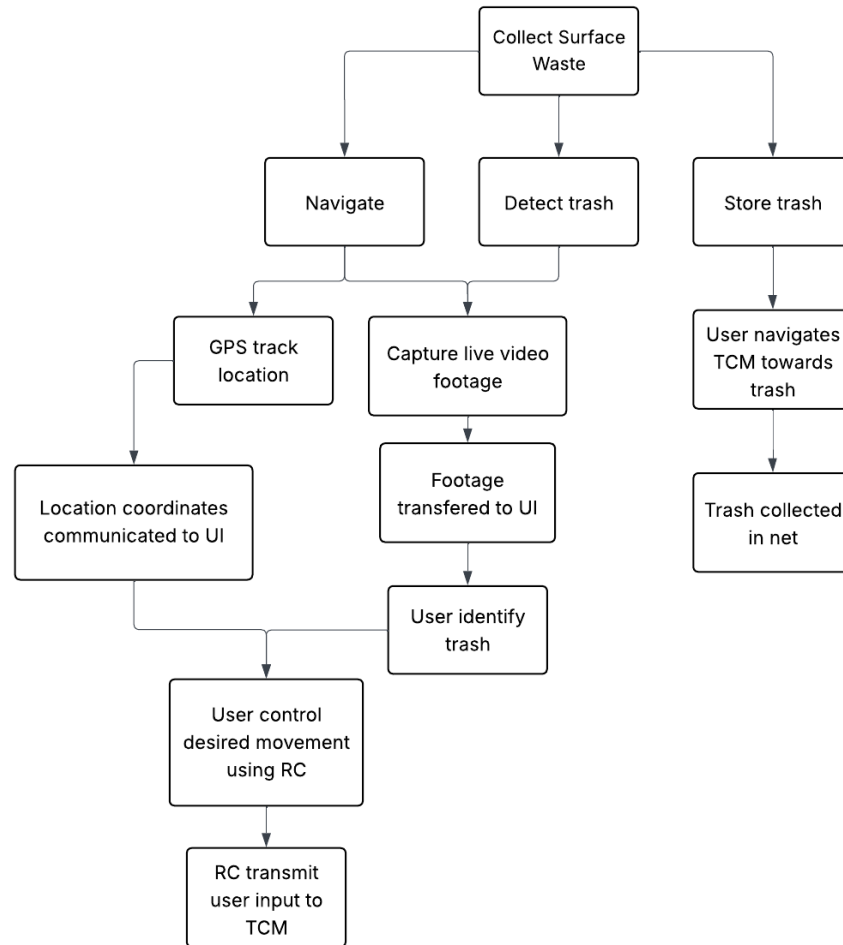
Constants with Nominal Values & Ranges

Table 3: Constants with Nominal Values & Ranges

Constants	Nominal Values & Ranges	Units
Maximum Distance from User	The minimum of either the RC range or the Wi-Fi range of the microcontroller.	meters [m]
Maximum Speed of Motors	TBD ; based on motor specifications and testing.	RPM [revolutions/minute]
Maximum Weight on System	TBD ; based on testing.	grams [g]
Maximum Transmission Delay (RC)	Maximum time delay between transmission of RC controller data and receipt of said data by the receiver (on-boat).	microseconds [us]
Maximum Transmission Delay (MCU)	Maximum time delay between collection of data from GPS and camera to receipt of data by the host PC (user).	microseconds [us]

Functional Decomposition

Figure 3: Functional Decomposition diagram



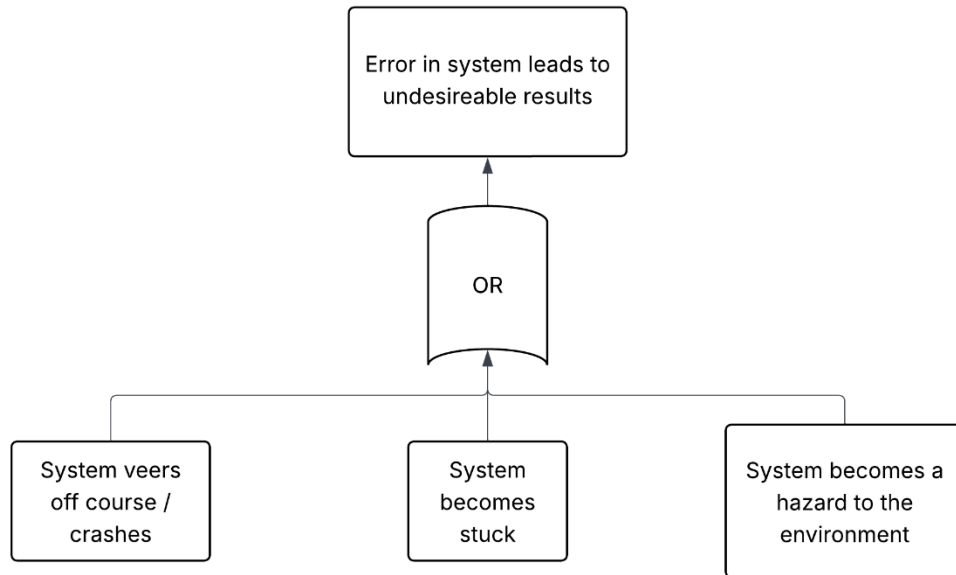
System Hazards

Critical Assumptions

1. The system is being operated by a person with the required competence who is familiar with the UM.
2. The system is being used in its intended environment and weather conditions without any strong wind gusts.
3. The system is being used in the appropriate body of water with appropriate water depth and minimal to no waves, such as a pond.
4. There are no motorized vehicles or human beings in the body of water.
5. The system's battery is fully charged, camera lens is clean, and there is no damage prior to deployment onto the water.
6. The TCM is configured correctly and empty prior to deployment onto the water.

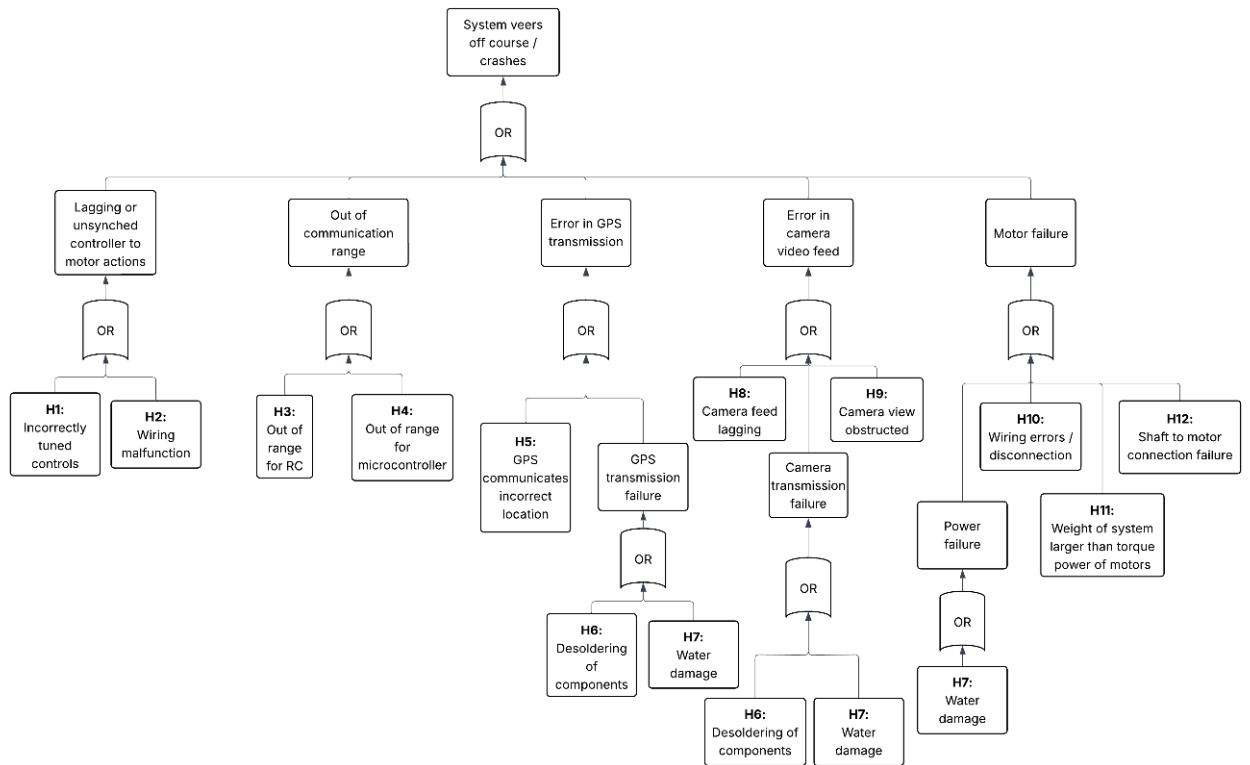
Hazards Analysis – Fault Tree Analysis

Figure 4: Top Level Hazards



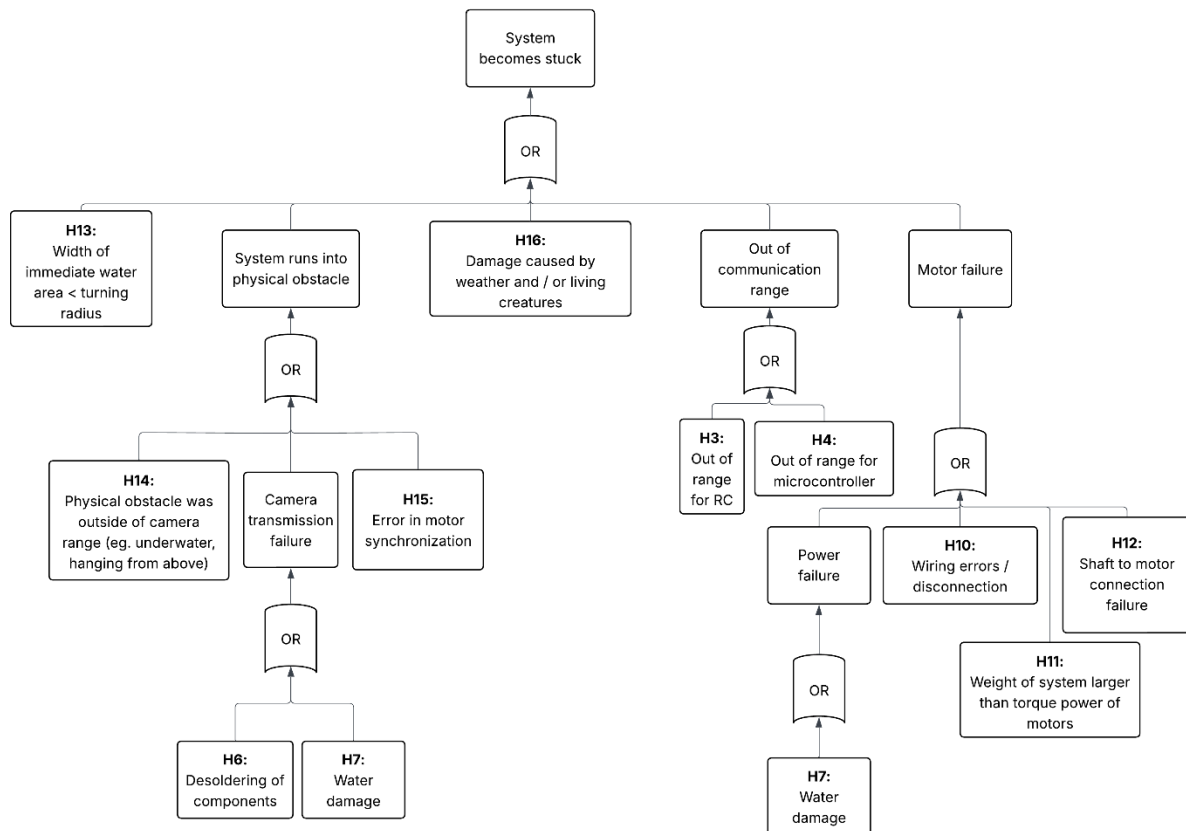
System Requirements & Hazards Analysis

Figure 5: Second Level Hazards 1



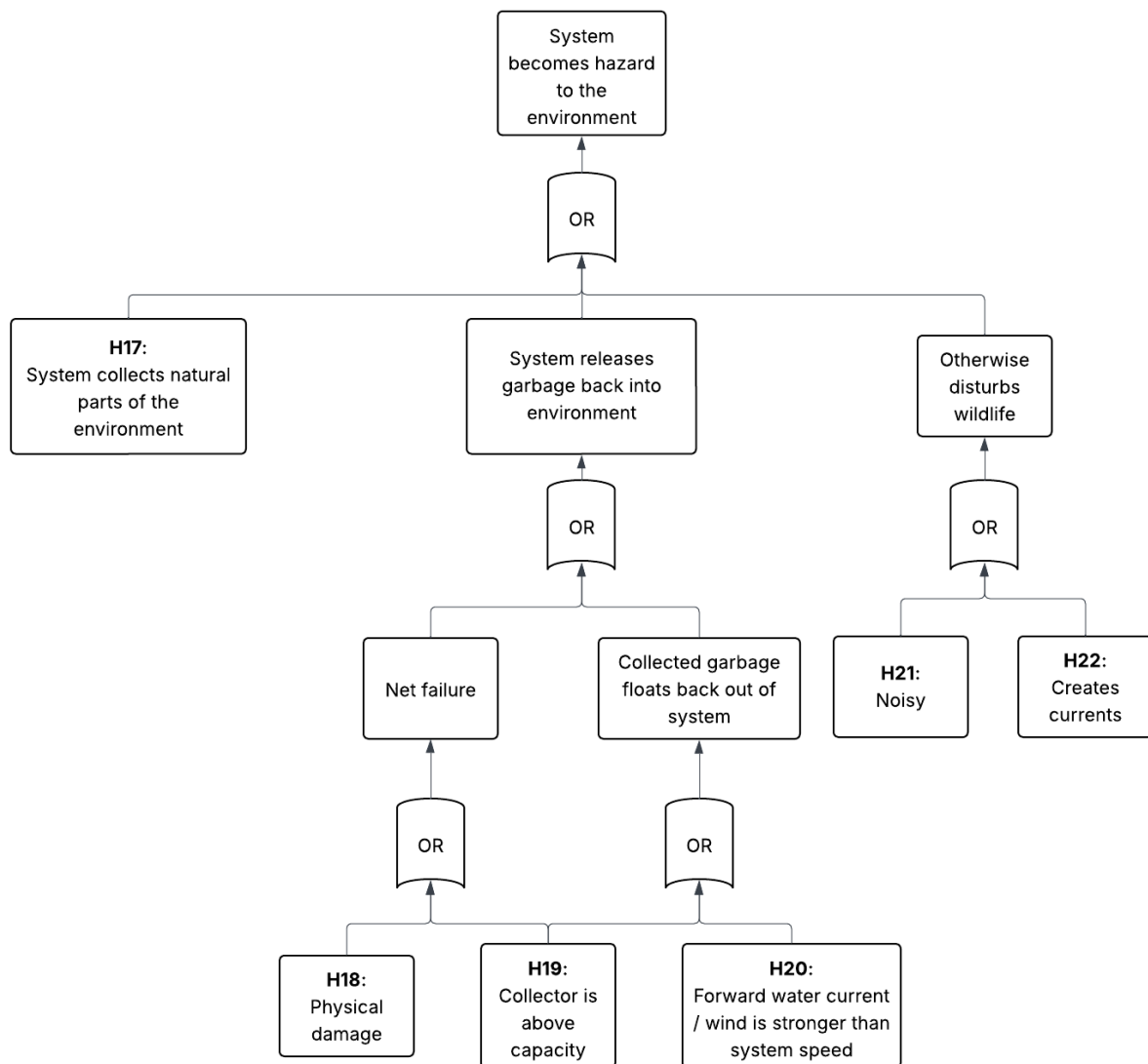
System Requirements & Hazards Analysis

Figure 6: Second Level Hazards 2



System Requirements & Hazards Analysis

Figure 7: Second Level Hazards 3



Risk & Mitigation Strategies

Table 44: Risk & Mitigation Strategies

HID	Risk Type	Description of Risk	Mitigation Strategy
H1	Incorrectly tuned controls	Incorrect tuning of the control system can lead to unpredictable behavior of the system such as failure to respond to control input as needed.	Rigorous tuning of controls during development process. Consider all edge cases.
H2 H15	Motor synchronization error	The motor being unsynchronized with the RC controller may cause loss of directional control.	Test motors beforehand to find the minimum distance at which synchronization with the RC is lost. This strategy is extracted from the RC's user manual.
H3	Out of RC range	If the system travels out of the bounds of the RC range this will lead to loss of signal thus the user will no longer be able to direct the system.	Motor synchronization errors are expected to only occur when out of range. So, strategy used to address motor synchronization also addresses this hazard.
H4	Out of MCU range	If the MCU is too far from the hosting server communication may be lost, causing control delays and the system being disconnected from the server.	Testing with MCU (client) and host PC (host), and consult MCU specifications, to find the minimum distance at which synchronization is lost. Indicate this distance as the absolute maximum range of the TCM in user manual.
H5	GPS communicates incorrect info	The GPS module could output delayed position data, which can lead to navigation errors.	Test and check all relevant and possible edge cases to guide development of GPS handling algorithm.
H6	Desoldering of components	Mechanical stress, heat, or poor assembly can cause electrical components to desolder, leading to a system failure.	Ensure the components are soldered properly prior to TCM assembly.
H7	Water damage to electrical components	Exposure to water may damage sensors, wiring, or the MCU, which can lead to system failure.	In user manual, include a warning that electrical components could be damaged. Use a waterproof casing for electrical components which will be evaluated to ensure waterproofness before placing electronics in casing.

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H8	Camera feed lagging	Delayed camera feed or signal loss can affect the operator's ability to see and control the robot accurately.	Test extensively beforehand, and consult camera and MCU specifications, to find maximum delay in camera transmission.
H9	Camera view obstructed	External objects may block the camera's view, reducing situational awareness and increasing collision risk.	In user manual, include a warning of this possibility and direct users to be attentive of potential obstructions during operation.
H7 H10	Motor power failure	The motors may stop functioning because of electrical faults, mechanical failure, or overload which will lead to the TCM stalling or immobility.	If GPS detects no change in position of TCM for 30 seconds during operation, system will give user a warning that motors are not functional.
H10	Disconnected wires	Improper assembly may cause wires to disconnect, preventing power delivery or signal transmissions to critical components.	Ensure wires are well-secured during assembly.
H11	TCM mass > motor torque power	If the total weight of the TCM exceeds what the motors can handle, the robot may fail to maneuver properly.	Verify during testing that the total TCM mass does not exceed the motors' rated torque output. Adjust design by using lighter materials if necessary to stay within safety limits.
H12	Shaft to motor connection failure	The connection between the motor and propeller shaft may loosen or break, resulting in loss of propulsion on one or both sides.	Secure motor well to propeller shaft and warn in user manual of risk of detachment.
H13	Water area width < turning radius	The TCM may be operated in an area too narrow for proper turning, increasing the likelihood of getting stuck.	Include risk in user manual and warn users to be cautious of this possibility.
H14	Physical obstacle outside camera range	Objects not visible in the camera's field of view may pose a collision risk.	Include risk in user manual and instruct users to operate the TCM in an area which does not have many low-overhanging branches.
H17	TCM collecting natural environment	TCM may collect natural material that are a part of the ecosystem reducing capacity and affecting system performance.	Include risk in user manual and warn users to be cautious of this possibility.
H16 H18	Physical damage to net	The collection net may tear from sharp debris or excessive load, leading to loss of collected waste.	Include risk in user manual and warn users to be cautious of this possibility.

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H19	Net above capacity	Overloading the net may reduce maneuverability, strain the motors, or cause the net to tear or detach.	Include net's capacity in user manual and warn users to use their best judgement during operation to ensure the net is regularly emptied so as to not exceed capacity.
H20	Forward water current/wind stronger than TCM speed	Strong currents or winds may push the TCM off course, reducing control.	Include warning in user manual not to operate in extreme weather conditions.
H21	Noisy	The motors or mechanical components may produce excessive noise, potentially disturbing wildlife or violating noise expectations.	Use sound-insulating materials around noisy areas of the TCM internals to muffle noise.
H22	TCM creating currents	The TCM's propulsion may generate disruptive water currents that affect its stability or environmental conditions.	Limit maximum operating speed of TCM and include in user manual.

System Requirements

Likelihood of Changing: How probable the requirement will be revised as development progresses. Below are the definitions of the low, medium, high likelihood of changing and will be added to all requirements tables below.

Table 5:5 Definition of Requirement's Likelihood of Change

Low	The requirement is very stable, meaning it is highly unlikely to change.
Medium	The requirement is somewhat stable, meaning minor adjustments may occur.
High	The requirement is unstable meaning it is most likely going to change.

Safety & Security Requirements Derived from Hazards

The following requirements identifies key failure modes such as loss of control, navigation errors, communication failures, and external environmental hazards. They are derived from the root causes of hazards and are classified as Safety & Security Requirements (SSR).

Table 66: Safety & Security Requirements

RID	HID	Description	Likelihood of Changing
SSR1	H7	Waterproof system: The electrical components in the TCM shall have waterproof housings to prevent damage from splashes.	Low
SSR2	H7 H10 H11 H12	Emergency stop: If TCM loses connection with RC for more than TBD Seconds, power supply shall be cut off.	Low
SSR3	H14	Alert for Environmental interference: TCM should alert user when motor is moving but GPS coordinates are not updating. <i>Rationale: Allows user to check for environmental interference (animals / branches / rocks / stuck / etc.)</i>	Medium
SSR4	H3 H4	Communication Range: TCM should alert user through UI when TCM is within TBD meter away from out-of-range boundaries.	Medium
SSR5		User Manual: User manual shall be created to prevent operational hazards.	Low
SSR5.1	H20	<i>User must operate system in ideal weather and water conditions.</i>	Low
SSR5.2	H9	<i>User must ensure that camera is clean/ view is not obstructed before operation.</i>	Low
SSR5.3	H17	<i>User must monitor the camera footage rigorously during operation.</i>	Low
SSR5.4	H16 H18	<i>User must check system upon recollection for physical damage.</i>	Low
SSR5.5	H19	<i>User must be wary of the amount of trash collected by the system to mitigate overflow.</i>	Low

System Requirements & Hazards Analysis

SSR5.6	H13	<i>User must be aware of max turning radius.</i>	<i>Low</i>
SSR6	H22	Speed limit: Maximum operating speed of TCM shall not exceed TBD m/s.	High
SSR7	H21	Sound insulation: System should be sound insulated as to not disturb the natural environment.	Medium
SSR8	H1 H2 H5 H6 H8 H10 H15	Component validation: All electrical components of system shall be tested prior to system assembly to validate specifications.	Low

Requirements Specification

The following requirements are the high-level behavioral requirements of the system.

Table 7: Behavioral Requirements

RID	Description	Likelihood of Changing
R1	Flotation: The TCM shall be floating during the entirety of the operation cycle.	Low
R2	Trash Collection: The TCM shall securely collect trash floating on the surface of small bodies of water.	Low
R3	Optimal Operation: The TCM shall operate reliably on the surface of water carrying all associated components and any collected trash up to TBD cm ³ .	Low
R4	System Movement: The TCM's movement should be remotely controlled by the user.	Medium
R5	Turning Radius: The system should allow forward movement and changes of direction with a TBD cm turning radius.	Medium
R6	Location Detection: The TCM component should be remotely located by the user using GPS coordinates.	Medium
R7	Visualization: The system shall provide the user with live video footage of the front view of the TCM.	Low

Performance Timing Requirements

The following requirements define the performance expectations of our surface-water trash collector, specifying how quickly the system must detect, collect, and dispose of floating debris in West Pond.

Table 8:8 Performance Timing Requirements

RID	Description	Likelihood of Changing
R8	Detection Response Time: Using camera visualization, the user should be able to detect a standard floating object approaching the TCM within TBD seconds.	Low
R9	Collection Cycle Time Per Item: The TCM should complete a full collection time under normal conditions (user detect → align → capture → store) for a properly sized item in less than TBD .	Medium
R10	Battery Efficiency Cycle: The system shall operate continuously for at least TBD hours before it needs to be recharged.	Low
R11	Alignment Time: The TCM should properly align the net to intercept an object within TBD sec from TBD meters away from the net.	Medium
R12	Direction Change Time: The TCM should be able to change direction by TBD degree angle in TBD seconds.	Low
R13	Thrust to Drag Response Time: The propulsion mechanism shall generate sufficient thrust to overcome drag and provide forward motion within TBD seconds of motor activation from rest. <i>Rationale:</i> This ensures that the system can overcome water and collected trash resistance quickly, preventing stall at low speeds.	Low
R14	Motor Response Time: Each of the motors should reach TBD of the user input speed within TBD seconds of receiving the control signal.	Low
R15	GPS Response Time: The delay between receiving GPS data from the BN-220 and the ESP32 main control loop processing shall be ≤ TBD	Low

Requirements for Hardware Environment

These requirements below define the hardware requirements for the surface-water trash collector, specifying the physical components and equipment needed for the system to function properly.

Table 99: Hardware Requirements

RID	Description	Likelihood of Changing
R16	Operation Temperature: The system should operate reliably between TBD degrees Celsius.	Medium
R17	UV-Stabilization: The TCM's plastic enclosures, plastic structure, and net must be UV-stabilized to limit degradation.	Low
R18	Operation Levels: The system's safe operation is guaranteed up to wind $\leq$ TBD , wave height $\leq$ TBD , current $\leq$ TBD m/s.	High
R19	Dynamic Load Operation: Net frame and attachments shall withstand the impact of dynamic loads (drag force) and hold up to TBD kg continuous load in the TCM.	Low
R20	Maintenance Accessibility: The system's compartments (battery, electronics, motors, GPS tracker, camera) shall be accessible for inspection/maintenance.	Low
R21	Component Mounting: BN-220 GPS Tracker and camera shall be mounted on a raised, unobstructed mount.	Low
R22	Camera Specifications: Camera viewport shall be optically clear. Camera must support required resolution for detection.	Low
R23	Trash Loss Prevention: The TCM should be in continuous forward motion to prevent the loss of trash collected.	Medium
R24	Connection Status: The system shall have sufficient wireless connection between the ESP32 and the controller.	Low

Requirements for Software Environment

The requirements below define the software requirements for the surface-water trash collector, specifying the microcontroller and the UI.

Table 1010: Software Requirements

RID	Description	Likelihood of Changing
R25	Real-time Control: The remote controller shall enable real-time control of the TCM's movement, allowing independent operation of the two motors.	Low
R26	Sensor Integration: The software shall interface with the onboard GPS module and live camera to collect position and visual data.	Low
R27	Data Communication: The system shall transmit data from the microcontroller to the external UI.	Low
R28	UI Support: The UI should display real-time TCM coordinates and live video feed.	Medium
R29	Synchronization and Timing: The software shall ensure synchronized operation between remote controller data, motor control, and data transmission.	Low
R30	Modularity: The software shall be structured in modules (motor control, GPS, camera, UI) to simplify development, testing, and maintenance.	Low
R31	Extensibility: The software environment should allow integration of additional modules or functionalities such as mapping, analytics, or remote monitoring.	High

List of Requirements That Are Likely to Change

All requirements that have *High* or *Medium* Likelihood of Changing are likely to change. Most of the requirements are shown to have values that are undetermined which are subject to change as we continue to proceed with the project. These values will be determined during the testing phase and are subject to change. For example, the performance timing requirements are the most likely to change because they depend on environmental factors, such as water conditions, battery levels, and trash load. Variations in weight, drag, and available thrust during testing can affect system response, making these requirements more sensitive and subject to adjustment. However, the format of these associated requirements will most likely not change.

In addition, if it is decided that navigation control will be centralized within the UI, rather than controlled by the RC, it is likely that additional requirements related to motor failure detection will be added.

List of Requirements That Are Not Likely to Change

All requirements that have *Low* Likelihood of Changing are not likely to change. The requirements that are unlikely to change are those without a defined value range such as ensuring the camera is mounted without obstruction, which is an obvious necessity, and that the boat remains continuously afloat. These are requirements that will need to be implemented to ensure our system successfully collects trash.